

COMPUTER NETWORKS 1

LAB 1: NETWORK DEVICES

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I. Objectives

- Get to know basic network devices.
- Understand functions of network devices.
- Able to connect different network devices together to form a simple network.

II. Content

1. Get to know network devices

- | | |
|--|---|
| ☒ Network Interface Card (NIC) | ☒ Routers |
| ☒ Cables | ☒ Access Points |
| ☒ Hub | ☒ Modems |
| ☒ Switches | |

2. Understanding functions of network devices

a. Network Interface Card (NIC)



NIC functions: Connects a computer to a network and sends and receives data.

Code of NIC processors: Ethernet controller / NIC chipset.

Check NIC of a computer, what is its MAC address? *74-D4-DD-6A-86-8F*

Cable to connect NIC to a network:

- **Type:** Twisted-pair cable (UTP)
- **Standard:** IEEE 802.3 Ethernet

b. Hubs



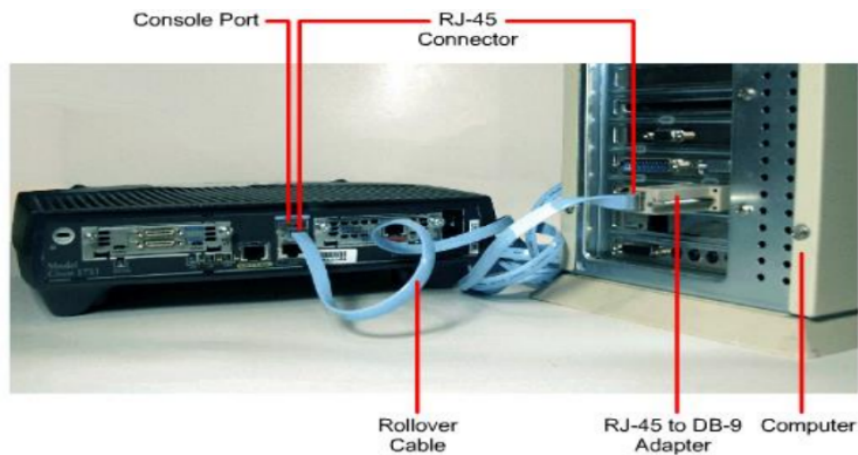
- **Roles of hub in a network:** Connects multiple devices in a LAN and forwards incoming data to all ports.
- **Main characteristics:** Operates at Layer 1 (Physical layer) and broadcasts incoming data to all ports.
- **Weaknesses of hub:** Low efficiency, more collisions, and less secure than switches.
- **Hub ports:** Usually 4, 8, 16, or 24 ports.

c. Switches



- **Roles of switches in a network:** Connects multiple devices in a LAN and forwards data to the correct destination.
- **Main characteristics of switches:** Operates at Layer 2 (Data Link layer) and uses MAC addresses to forward frames.
- **Differences between hubs and switches:** A hub broadcasts data to all ports, while a switch forwards data only to the appropriate destination port.
- **Weaknesses of switches:** More expensive and more complex than hubs; performance can decrease when overloaded.
- **Switch ports:** Usually 8, 16, 24, 48, or more ports.

d. Routers



- **Roles of routers in a network:** Connects different networks and forwards packets between them.
- **Main characteristics of routers:** Operates mainly at Layer 3 (Network layer) and uses IP addresses to route packets.
- **Differences between routers and switches:** A router connects different networks, while a switch connects devices within the same network.
- **Router ports:** Usually has WAN and LAN ports; it may also have a Console port.

e. Access Points



- **Roles of access points:** Provides wireless access to a wired network and connects wireless devices to the network.
- **Main characteristics of access points:** Provides Wi-Fi connectivity and operates mainly at Layer 2.
- **Access point's interfaces:** Wireless interface and Ethernet (RJ-45) interface.
- **Compare access point and other networking devices mentioned above:** An access point connects wireless devices to a network, while a hub broadcasts data to all ports, a switch forwards frames using MAC addresses, and a router forwards packets between different networks.

f. Modem



Differentiate and describe roles and characteristics for each type of modem:

- **Dial-up modem:** Uses a telephone line to establish an Internet connection by dialing a number. It has a low transmission speed and normally cannot be used for voice calls and Internet access simultaneously.
- **ADSL Modem:** Uses a telephone line to provide high-speed Internet access. It allows voice calls and Internet access to be used simultaneously.
- **Cable Modem:** Uses a cable television network to provide high-speed Internet access. The available bandwidth is shared among users in the same area.

3. Connecting network devices

Identify the type of network cable that can be used for the following network connections:

- Computer and hub: **Straight-through cable (Cáp thẳng)**
- Computer and switch: **Straight-through cable (Cáp thẳng)**
- Computer and router: **Crossover cable (Cáp chéo)**
- Hub and hub: **Crossover cable (Cáp chéo)**
- Hub and switch: **Crossover cable (Cáp chéo)**
- Hub and router: **Straight-through cable (Cáp thẳng)**
- Switch and switch: **Crossover cable (Cáp chéo)**
- Switch and router: **Straight-through cable (Cáp thẳng)**
- Router and router: **Crossover cable (Cáp chéo)**